Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/18/2014
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Charged Particle-Induced Autosomal Mutagenesis in Murine Tissue and Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	03/14/2012	**End Date:**	03/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2016 per S. Monk/LaRC (Ed., 9/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Turker, Mitchell (Oregon Health & Science University) Grygoryev, Dmytro (Oregon Health & Science University) Grossi, Gianfranco (Universita Federico II) Gauny, Stacey (Lawrence Berkeley National Laboratory) Ohlrich, Anna (Oregon Health & Science University) Hryciw, Gwen (Oregon Health & Science University)		
Grant/Contract No.:	NNJ12HB88I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts in space are exposed to a variety of charged particle radiations, some of which travel at very high velocities that can penetrate spacecraft, damage tissues, and increase the risk of cancer. This proposal addresses two cancer development processes, genomic instability and aneuploidy, links them to each other and to specific mechanisms of mutagenesis at an endogenous autosomal locus in epithelial cells. The aims proposed use a comparative biology approach and a transgenic mouse model that involves both in vitro and in situ approaches. This strategy enables quantitative, coupled analyses of early changes in cells that survive the exposure and a direct test of the relevance of these changes to the development of persistent changes in situ. The studies address the radiation quality dependence of genomic instability and aneuploidy focusing on charged particle beams spanning a critical LET range from 25 keV/µm to 151 keV/µm, and will define the shape of the dose-response for these endpoints, particularly at low fluence. The rationale for working in this range of LET is the observation of increased biological effectiveness for cancer relevant endpoints in our previous studies, including mutagenesis in human and rodent cells, and work by others concerning Harderian gland tumorigenesis in mice. The investigations test the hypothesis that the induction of genomic instability and aneuploidy will increase as a function of ionization density for low fluence exposures, leading to genome-wide alterations in viable, mutant cells found in normal tissue. This project will define links between the processes that lead to early establishment of genomic instability, aneuploidy, and mechanisms of autosomal mutagenesis found in many kinds of human epithelial cancers. These quantitative, mechanism-based studies will aid in the extrapolation of results across species to help reduce the uncertainties in cancer risk projections for astronauts exposed to spaceflight radiations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research examines the effects of charged particle radiations with different ionization densities using normal mammalian epithelium. The research is designed to help determine links between cancer development processes at relatively low doses. This work is of relevance to naturally occurring Earth-based exposures to densely ionizing radiations such as radon. It will also inform our understanding of the mechanisms underlying potential risks to normal tissues associated with the implementation of charged particle radiation therapy.
Task Progress:	This project was initiated in Spring 2012. During the second year of funding, we conducted replicate experiments to determine the dose-response for autosomal mutagenesis following exposure to graded doses of 250 MeV/amu O ions (LET = 25 keV/µm) using cultured kidney cells from B6D2F1 mice. A dose-dependent increase in Aprt mutations was noted for B6D2F1 Aprt +/- cells exposed in Spring 2013 at the NSRL accelerator. Studies begun in year 1 with B6D2F1 Aprt +/- mice exposed to 263 MeV/amu Si ions (LET = 78 keV/µm at the midline) at the NSRL were completed. This work demonstrated that the linear dose-response for Si ion-induced mutations in B6D2F1 Aprt +/- cells exposed to Si ions of the same LET was conserved if the kidney epithelial cells were exposed in situ and harvested 3-5 months later. Dose-dependent cytotoxicity was also noted for this cohort. Additional studies with B6D2F1 Aprt +/- mice were initiated in Fall 2013 with a separate cohort of mice exposed to 250 MeV/amu O ions. These studies are ongoing. We completed the analysis of the types of molecular changes associated with the etiology of Si ion-induced Aprt mutations (240 MeV/amu, LET= 78 keV/µm). The goal is to identify signatures of exposure that may support our central hypothesis that links together several cancer development processes including centrosome abnormalities, aneuploidy, and molecular mechanisms of mutagenesis. In cultured cells, the predominant classes of mutations observed among 58 mutant clones isolated after exposure to a benchmark dose of 1.6 Gy were apparent mitotic recombination events and deletions. The spectrum of Si ion-induced mutations in cultured kidney cells could be readily distinguished from spontaneously arising Aprt mutants derived from un-irradiated cultures, where intragenic events predominate. Our ongoing work indicates a link between certain types of mutational events on mouse chromosome 8 leading to Aprt mutations and additional genome-wide changes on other chromosomes (genomic LOH). Evidence for genomic LOH was found in the cultured kidney cells exposed to Si ions, indicating that Si ions of intermediate LET can produce these types of events in apparently normal epithelial cells at short times post-exposure. In this regard, mutants that arose after exposure of cultured cells to Si ions resembled mutants isolated after exposure to more densely ionizing Fe ions (LET=151 keV/µm), but not mutants isolated after exposure to sparsely ionizing protons (LET= 0.24 keV/µm). Similar analyses were performed on Si ion-induced mutants isolated from irradiated mice. Fifty-nine mutant clones isolated after exposure to a benchmark dose of 1.6 Gy also demonstrated an increase in apparent mitotic recombination and deletions when compared with the spontaneous mutant spectrum from non-irradiated kidneys. Taken together, these results indicate that Si ions of intermediate LET can produce large-scale genomic changes at short time post-irradiation that persist for many months in an apparently normal epithelium in vivo. During Spring and Fall 2013, two large series of replicate, independent cultures of B6D2F1 Aprt +/- cells were irradiated with lower doses (0.4 Gy or 0.8 Gy) of Si ions to collect both non-mutant and Aprt-deficient mutant clones for further characterization of the various cancer development processes described above. These cultures were archived for analysis. This work is complemented by the analysis of Aprt-deficient clones from mice exposed to lower doses (0.4 Gy or 0.8 Gy) of Si ions. Efforts are underway to complete a low dose analysis of Fe ion-induced mutants following exposures to a fluence corresponding to an average of one particle traversal/cell nucleus.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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